18 August 2026

Editorial Team
Scientific Reports

Re: Article submission, "Evidence-Qualified Alignment of Pathology Foundation Models with Morphologic, Molecular, and Outcome Axes in Prostate Cancer"

Dear Editors,

Please consider our manuscript, "Evidence-Qualified Alignment of Pathology Foundation Models with Morphologic, Molecular, and Outcome Axes in Prostate Cancer," for publication as an Article in *Scientific Reports*.

Pathology foundation models can encode clinically meaningful information, but recovery of a target from a representation does not establish that a downstream model uses that target or that the relationship will transport across cohorts. We address this problem by comparing six non-interchangeable prostate-cancer axes—grade/ISUP, tumor phenotype/content, PTEN loss, androgen-receptor activity, SPOP mutation, and recurrence—across frozen CONCH and Virchow representations. The study evaluates recoverability together with external transport, conditional information, site uncertainty, encoder and sampling sensitivity, endpoint fidelity, and a narrowly scoped internal functional-sensitivity analysis.

The results establish a deliberately qualified evidence hierarchy. Grade/ISUP and tumor phenotype/content show the strongest representation evidence; PTEN and androgen-receptor activity remain context-sensitive; SPOP is evaluated but unsupported in the frozen design; and recurrence depends on endpoint and reference specification. Targeted removal of an ISUP-correlated direction changes two internally locked biochemical-recurrence heads beyond matched-random controls, but the study does not claim indispensable mechanism, external functional transport, clinical increment, or improved clinical outcomes. We believe this combination of multi-axis evidence, explicit claim boundaries, and auditable provenance is well aligned with the broad methodological and biomedical readership of *Scientific Reports*.

The manuscript reports secondary analyses of deidentified data from previously published public or access-governed repositories. No new participants were recruited, and no identifiable participant information is reported. Original ethical approvals and consent or waivers remain those reported by the source studies and repositories. The authors declare no competing interests. The work was supported by NRF-2023R1A2C1006639 and in part by Gyeongsang National University.

Aggregate source tables, provenance registries, analysis-code snapshots, and deterministic figure-reproduction tests are available in the public release repository under the immutable release tag `v1.0.2-submission`. Reviewer suggestions or exclusions and the status of any prior discussion with a Scientific Reports Editorial Board Member will be declared by the corresponding author in the online submission system.

Thank you for considering our manuscript.

Sincerely,

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